# Forecasting electricity usage on 48h window

Oksana Yefimova

# The Key Lessons

1. Models that use the actual weather on past windows look much better than they really are.
2. On the test we only have a weather forecast, and its errors grow from day 1 to day 2.
3. So every comparison used forecast weather, or a simulation of forecast errors.
4. Day 1 uses the weather forecast. Day 2 repeats the last known weather.

# The final model

## SARIMA(3,0,0)(0,1,3) with a 24-hour season

Extra inputs: cloud cover, temperature, wind speed.

Trained on the last 7 days of history.

Interval: 95% quantiles of past errors, separately for day 1 and day 2, never below zero.

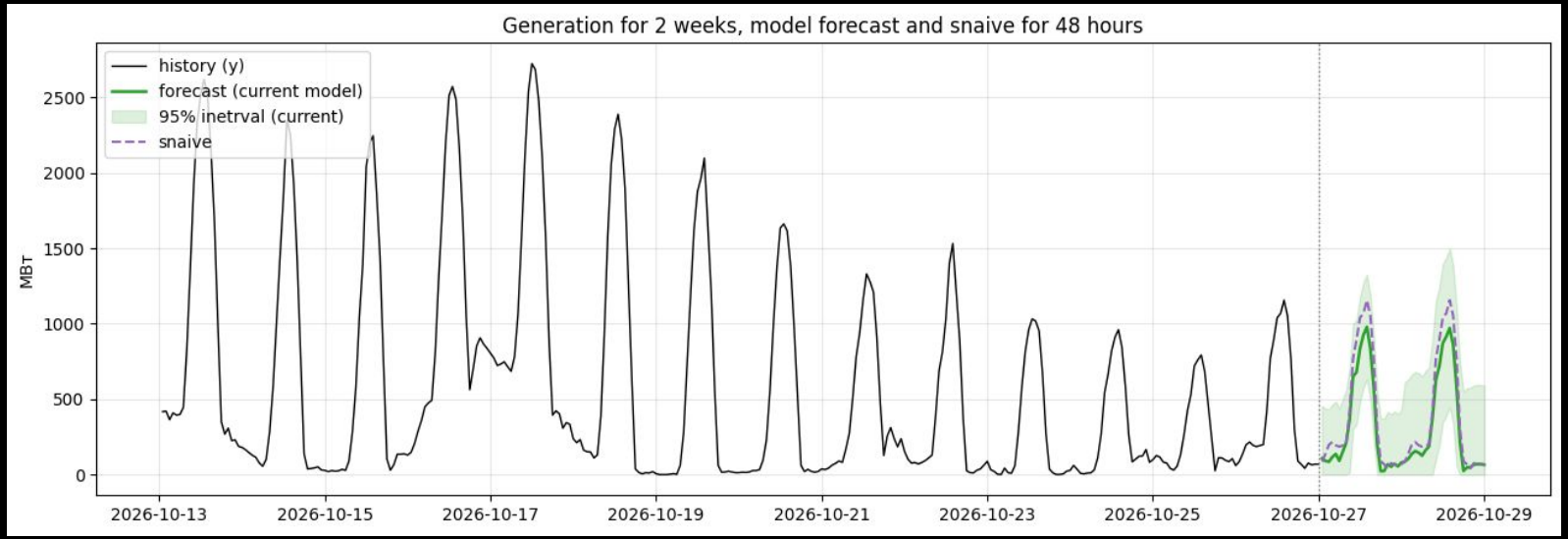
# Validation

RMSE and Winkler, benchmark vs final model, on 6 windows and on the 5 most recent.

| Model | Windows | RMSE | Winkler | Coverage |
|---|---|---|---|---|
| Benchmark | 6 | 267.5 | 1564 | 0.944 |
| Final model | 6 | 225.0 | 990 | 0.955 |
| Benchmark | 5 recent | 243.2 | 1181 | 0.979 |
| Final model | 5 recent | 214.6 | 970 | 0.962 |

6 test windows in total, the 5 most recent are the closest to the real test week.

# The forecast for 27–29 October



# Public Leaderboard

| | RMSE | MAPE | Winkler | Coverage | Score |
|---|---|---|---|---|---|
| My model | 83.1 | 53.2% | 529 | 100% | 11.8 |
| Benchmark | 115.7 | 84.3% | 521 | 100% | 14.6 |

Better than the benchmark on RMSE, MAPE and overall score

Slightly worse on Winkler, the interval is wider than needed this week